



Girijananda Chowdhury University Guwahati 781017, Assam

Unit Test - II

BCE23405T Structural Analysis

Session: 2024-2025

Batch: 2023-2027

Submission Date :: 23 April 2025

Total Marks: 30

Time : 1 hour.

Question 1

[CO2]

- If a unit rotation is applied at one end of the fixed beam, the moment produced at the other end of the beam will be (1 Marks)
- Sway occurs when the portal frames are in both geometry and loading. (1 Marks)
- Define column and strut. (1 Marks)
- A solid round bar 3 m long and 5 cm in diameter is used as a strut, determine the crippling load if one end hinged and another end fixed. Take $E = 2 \times 10^5 \text{ N/mm}^2$. (3 Marks)
- Derive the expression for crippling load for columns with one end fixed and another end hinged (4 Marks)

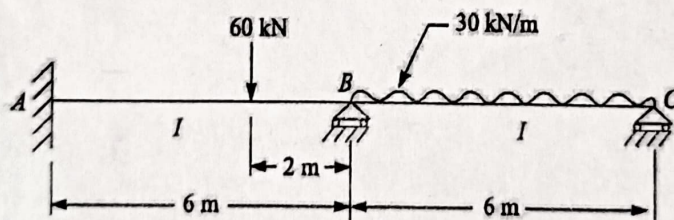
Question 2

[CO4]

- Analyse the continuous beam shown in figure below using Slope Deflection Method if support B sinks by 20 mm. Draw Shear Force and Bending Moment Diagrams.

$$E = 2 \times 10^5 \text{ N/mm}^2, I = 4 \times 10^7 \text{ mm}^4$$

(10 Marks)



OR

- Analyse the beam shown in Figure using Slope deflection method. Draw the shear force and bending moment diagrams for the beam. Take EI as constant.

(10 Marks)

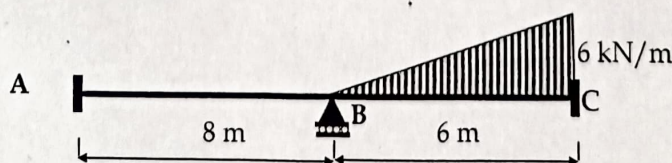


Figure Q2